

Dataset Description

Data Analysis Midterm Project | Malak Usama Mohamed

Dataset Title

Diabetes 130-US Hospitals for Years 1999–2008 (Diabetic Patient Readmission Dataset)

Dataset Source

The dataset originates from the Health Facts national data warehouse (Cerner Corporation, Kansas City, MO), which aggregates electronic health record data from hospitals across the United States. It was extracted and prepared for public research use by Strack et al. (2014) and published in *BioMed Research International* under the title “Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records.” The full dataset is publicly hosted on the UCI Machine Learning Repository (archive.ics.uci.edu, “Diabetes 130-US hospitals for years 1999–2008”). It reflects real, de-identified clinical encounters — not synthetic or generated data — drawn from 130 hospitals and integrated delivery networks across the Midwest, Northeast, South, and West regions of the U.S.

Business Domain

Healthcare — specifically, hospital operations and clinical quality analytics, focused on inpatient diabetes management and readmission performance.

Number of Rows and Columns

The original source file contains 101,766 encounters (rows) and 50 attributes (columns). For this project, the dataset was narrowed to 22 core variables most relevant to the business objective (patient demographics, admission and discharge details, clinical utilization metrics, diabetes-specific test results and medication indicators, and the readmission outcome). Three additional descriptive columns were engineered from existing numeric ID fields to improve readability, bringing the working file to 101,766 rows and 25 columns.

Description of Variables

Variable	Type	Description
encounter_id	Numeric	Unique identifier for each hospital encounter
patient_nbr	Numeric	Unique identifier for each patient (allows tracking repeat encounters)
race	Categorical	Patient race; missing values recoded as “Prefer Not to Say”
gender	Categorical	Patient gender (Female, Male, or Prefer Not to Say)
age	Categorical	Age grouped into 10-year intervals, e.g. [70-80)
admission_type_id / admission_type	Categorical	Numeric code and mapped label for how the patient was admitted (e.g. Emergency, Elective, Urgent)
discharge_disposition_id / discharge_disposition	Categorical	Numeric code and mapped label for discharge outcome (e.g. discharged home, transferred, expired)
admission_source_id / admission_source	Categorical	Numeric code and mapped label for referral source (e.g. Physician Referral, Emergency Room)
time_in_hospital	Numeric	Length of stay in days (1–14)

Variable	Type	Description
num_lab_procedures	Numeric	Number of lab tests performed during the encounter
num_procedures	Numeric	Number of non-lab procedures performed
num_medications	Numeric	Number of distinct medications administered
number_outpatient	Numeric	Outpatient visits by the patient in the year prior
number_emergency	Numeric	Emergency visits by the patient in the year prior
number_inpatient	Numeric	Inpatient visits by the patient in the year prior
number_diagnoses	Numeric	Total number of diagnoses recorded for the encounter
max_glu_serum	Categorical	Glucose serum test result (>200, >300, Norm, or None if not tested)
A1Cresult	Categorical	HbA1c test result (>7, >8, Norm, or None if not tested)
insulin	Categorical	Change in insulin dosage during the encounter (Up, Down, Steady, No)
change	Categorical	Whether any diabetes medication was changed (Ch / No)
diabetesMed	Categorical	Whether a diabetes medication was prescribed at all (Yes / No)
readmitted	Categorical (target)	Readmission outcome: <30 days, >30 days, or No readmission

Why I Selected This Dataset

I selected this dataset because it reflects genuine, large-scale hospital operations data rather than a synthetically generated one, which gives the analysis real credibility and lets variable relationships (e.g., between prior hospital utilization and readmission) reflect true clinical patterns rather than artifacts of a random data generator. The healthcare domain also aligns closely with my own professional background, having worked with CMS Medicare claims and insurance-related data in a healthcare-adjacent operations role, which gave me useful context for interpreting the variables and framing the business questions.

Business Problem or Analytical Objective

Hospital readmissions are a major cost and quality concern in U.S. healthcare, particularly for patients with diabetes, whose glycemic management during a hospital stay can meaningfully influence their risk of returning within 30 days. The analytical objective of this project is to identify which patient and encounter-level factors — such as prior hospital utilization, glucose/HbA1c testing, insulin dosage changes, age, and admission type — are most strongly associated with early readmission, and to translate those patterns into a KPI-driven dashboard that hospital administrators could use to target high-risk patients and evaluate performance against a readmission-rate benchmark.

Why the Dataset Satisfies the Project Requirements

- Structured and tabular: the data is organized in rows (encounters) and columns (attributes), suitable for direct use in Excel and Power BI.
- Real-world domain: falls squarely within Healthcare, one of the domains explicitly listed in the project brief.
- Meets minimum size requirements: 101,766 rows and 22–25 working columns, well above the 1,000-row / 8-column minimum.

- Numerical and categorical mix: numeric fields (time_in_hospital, num_medications, num_lab_procedures, etc.) sit alongside categorical fields (race, admission_type, insulin, readmitted), enabling grouping, filtering, and aggregation.
- Measurable business metric: the readmitted field and time_in_hospital both serve as concrete, measurable outcomes for KPI development.
- Genuine data quality issues: the raw data included inconsistent placeholder values ("?" in race), coded categorical fields requiring translation (admission/discharge/source IDs), and fields with meaningful non-response categories (max_glu_serum, A1Cresult) — all of which required real data cleaning and documented judgment calls rather than artificially introduced errors.
- Strong analysis potential: exploratory analysis revealed clear, meaningful patterns — most notably a roughly four-fold increase in 30-day readmission rate between patients with zero and six prior inpatient visits — providing a strong foundation for dashboard storytelling and business insight.